Numerical Analysis I

Quiz 12B Name:_____

Question 1.
Consider the following quadrature rule on $[a, b]$. Let

$$h = \frac{b-a}{3}, \qquad x_i = a + ih, \quad i = 0, 1, 2, 3.$$

The quadrature rule is given by

$$\int_a^b f(x)\,dx \approx \frac{3h}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right].$$

For $f \in C^4([a, b])$, the error is given by

$$E(f) = -\frac{3h^5}{80} f^{(4)}(\xi),$$

for some $\xi \in (a, b)$, where $E(f)$ denotes the exact integral minus the quadrature approximation.
This is called **Simpson's 3/8 rule**. This is the Closed Newton–Cotes rule with 4 nodes (3 subintervals).
Now let n be a positive integer, and redefine

$$h = \frac{b-a}{3n}, \qquad x_i = a + ih, \quad i = 0, \ldots, 3n.$$

1. Construct the composite quadrature rule for $n = 2$. Express your answer as a weighted sum of $f(x_0), f(x_1), \ldots, f(x_6)$.

2. Generalize your result to an arbitrary positive integer n. Express the composite quadrature rule $Q_n(f)$ as a weighted sum of function values.

3. Using the error formula for the basic rule, show that the error of the composite rule satisfies

$$\left| \int_a^b f(x)\,dx - Q_n(f) \right| \leq Ch^4,$$

where C is independent of n. Determine C explicitly in terms of a, b, and $\max_{x\in[a,b]} |f^{(4)}(x)|$.

4. If the step size h is halved, by what factor does the error bound decrease? Justify your answer.

Question 2.
Consider the following open quadrature rule on $[a, b]$. Let

$$h = \frac{b-a}{4}, \qquad x_i = a + ih, \quad i = 0, \ldots, 4.$$

The quadrature rule is given by

$$\int_a^b f(x)\,dx \approx \frac{4h}{3}\left[2f(x_1) - f(x_2) + 2f(x_3)\right].$$

1. Determine the degree of exactness of this quadrature rule. Justify your answer.
2. What do you expect the order of accuracy of this quadrature rule to be in terms of h? No proof is required.